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Christensen

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(54) **METHOD OF ASSEMBLY OF AN
ENDOSCOPE CONTROL SYSTEM**

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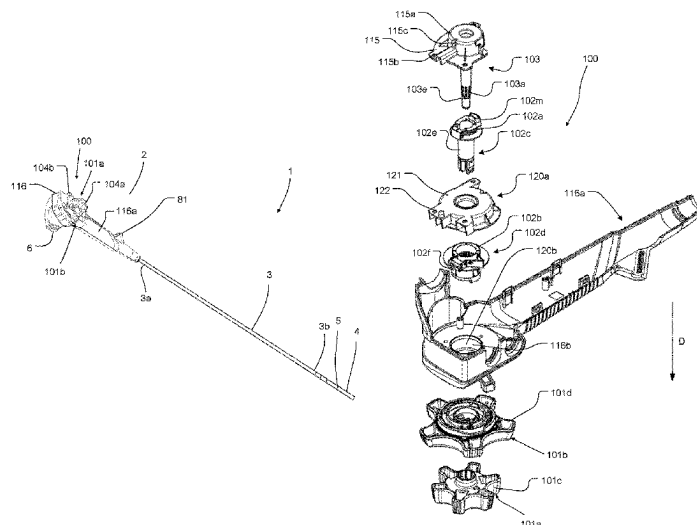
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(57) **ABSTRACT**

A method of assembly of an endoscope control system including: holding a first control wheel unit in a position; moving a second control wheel unit in an assembly direction to position the second control wheel unit on the first control wheel unit; moving a housing frame in the assembly direction to position the housing frame on the second control wheel unit; moving a second shaft unit in the assembly direction so that a second shaft is positioned to extend through the connection hole of the housing frame and snaps into engagement with the second control wheel unit; and moving a first shaft unit in the assembly direction so that a first shaft is positioned through the second shaft and snaps into engagement with the first control wheel unit.

13 Claims, 7 Drawing Sheets



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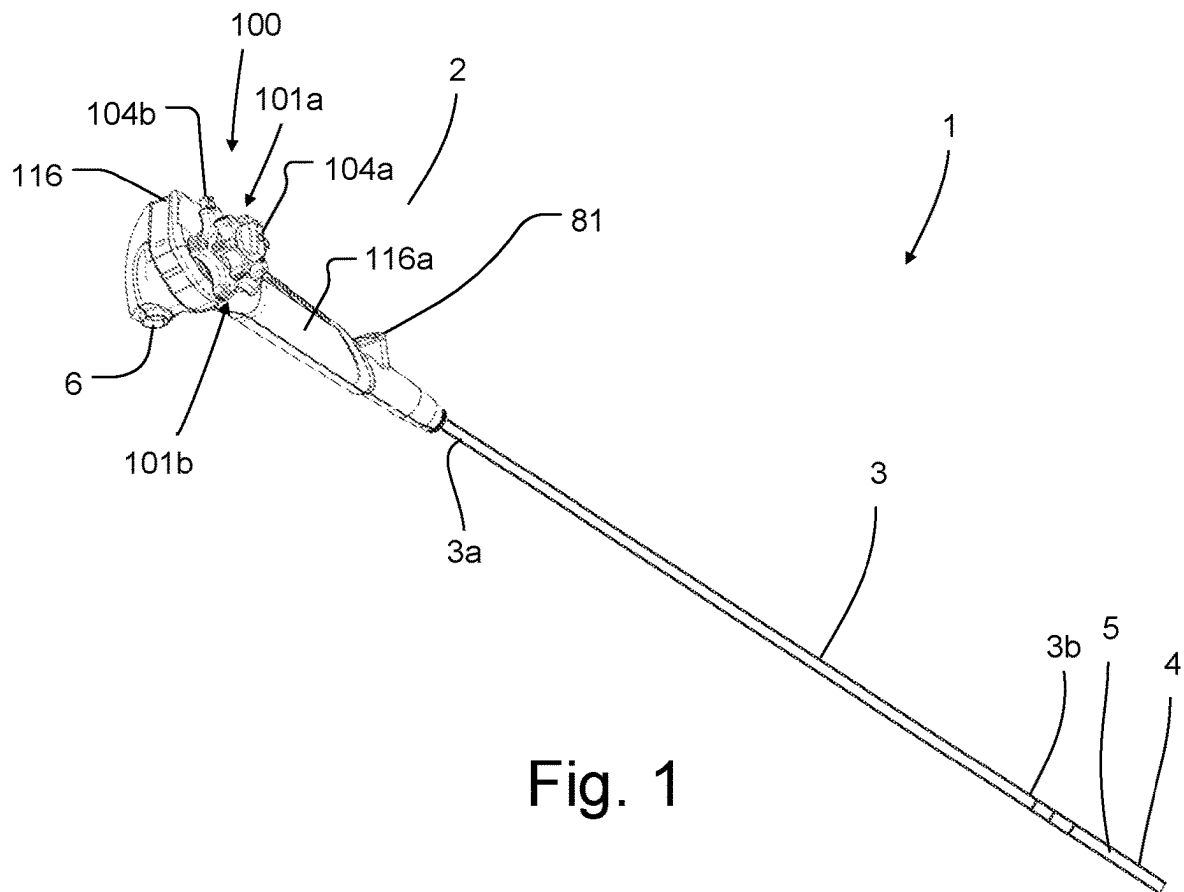
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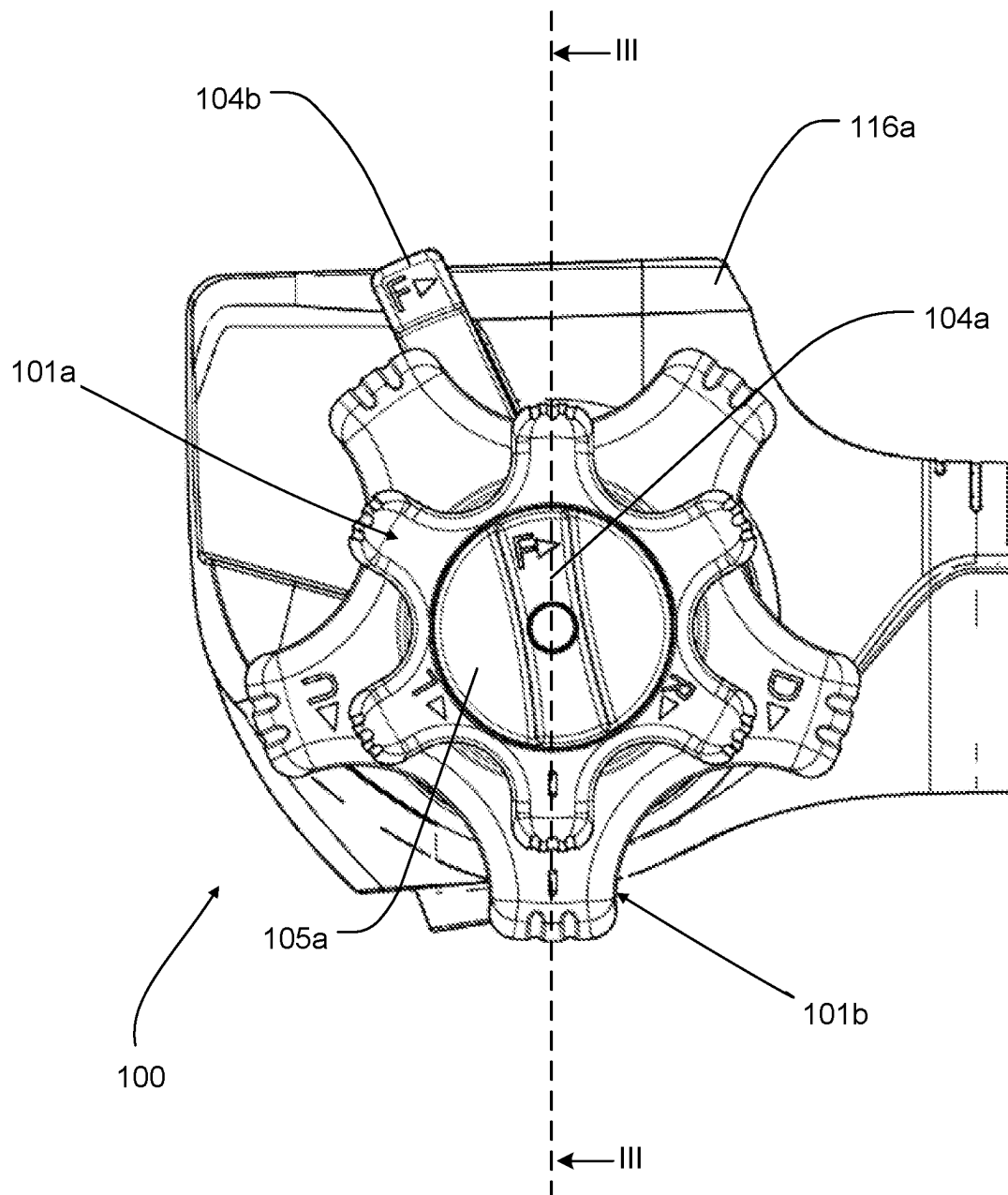


Fig. 2

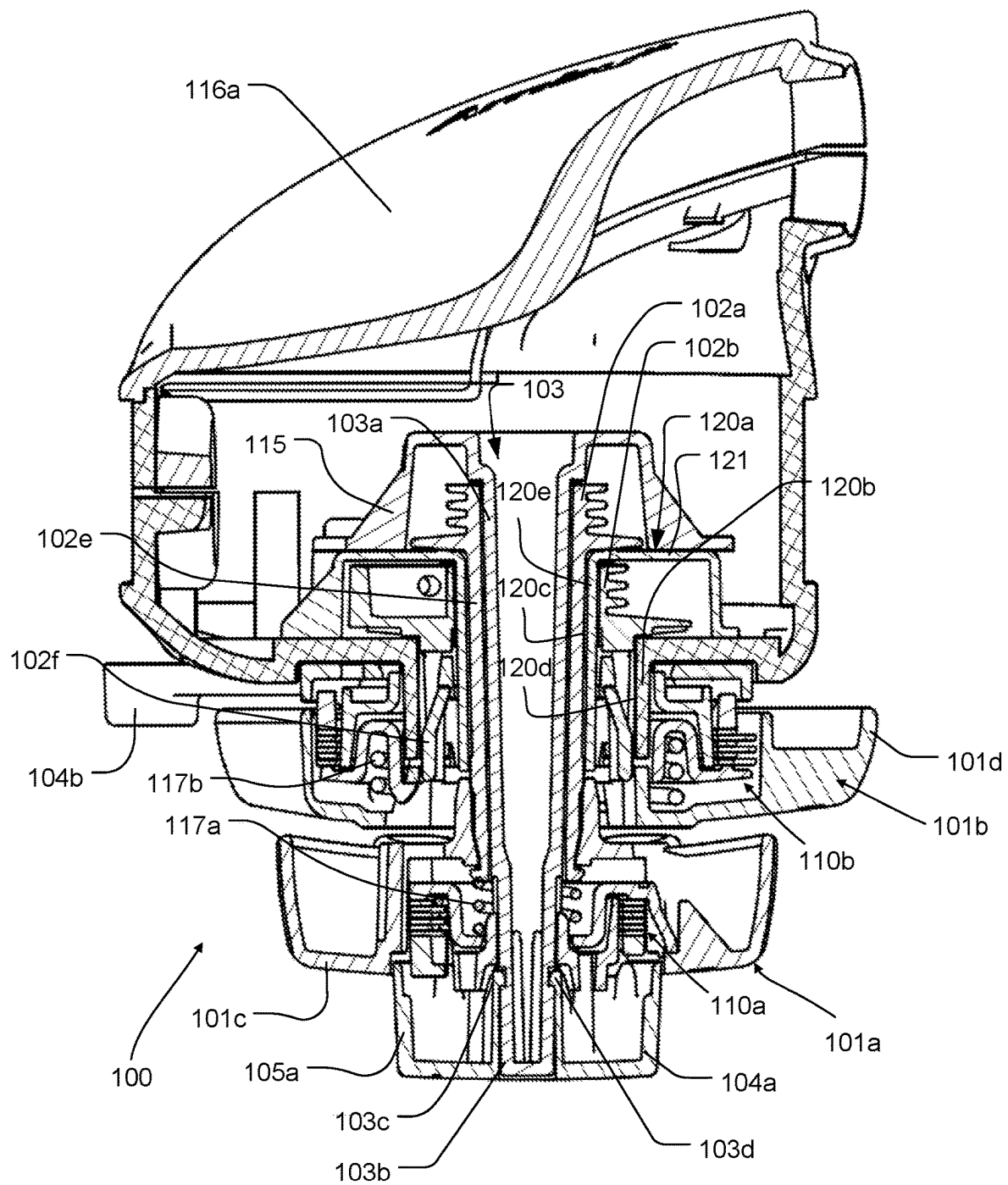


Fig. 3

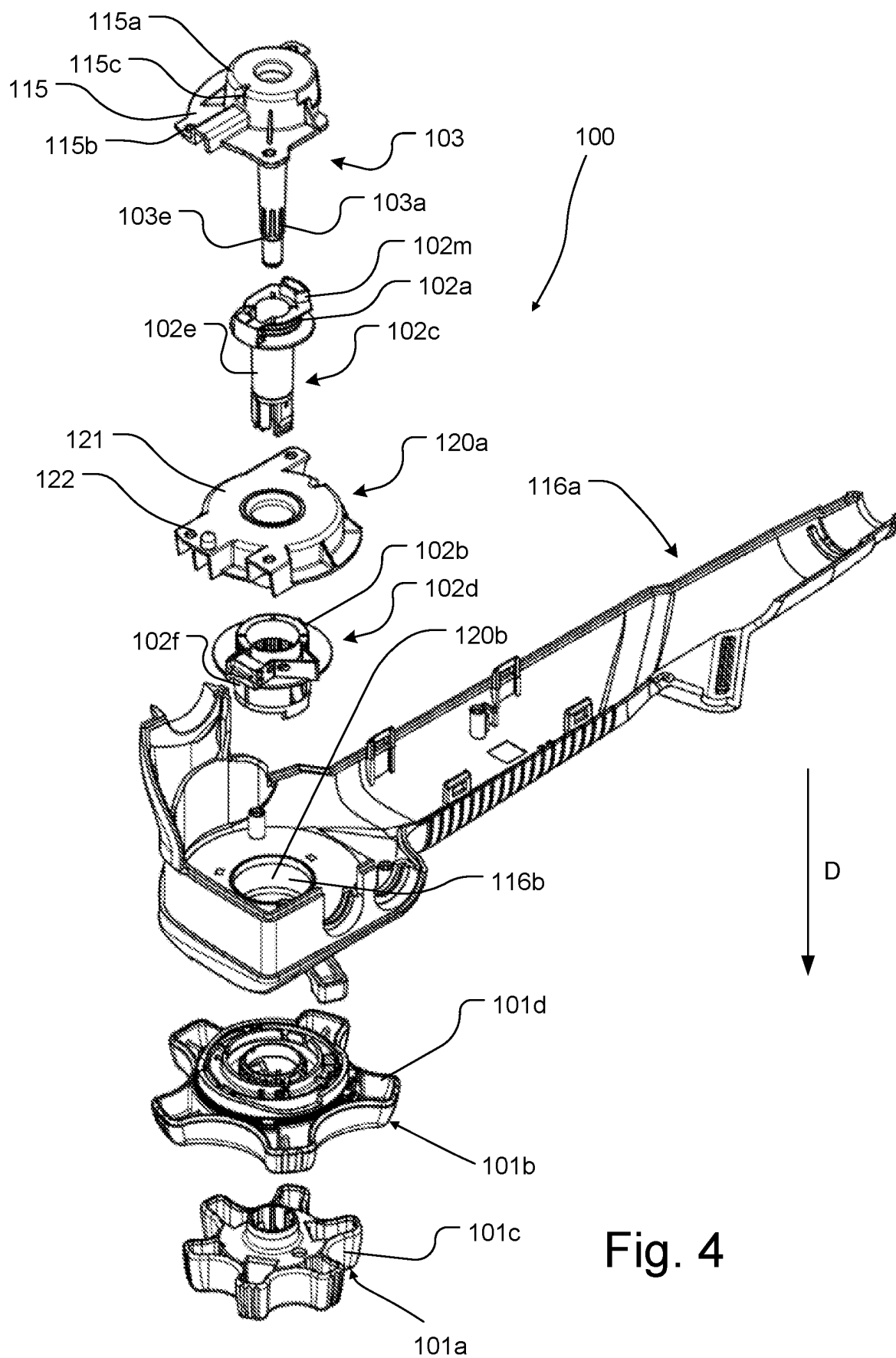
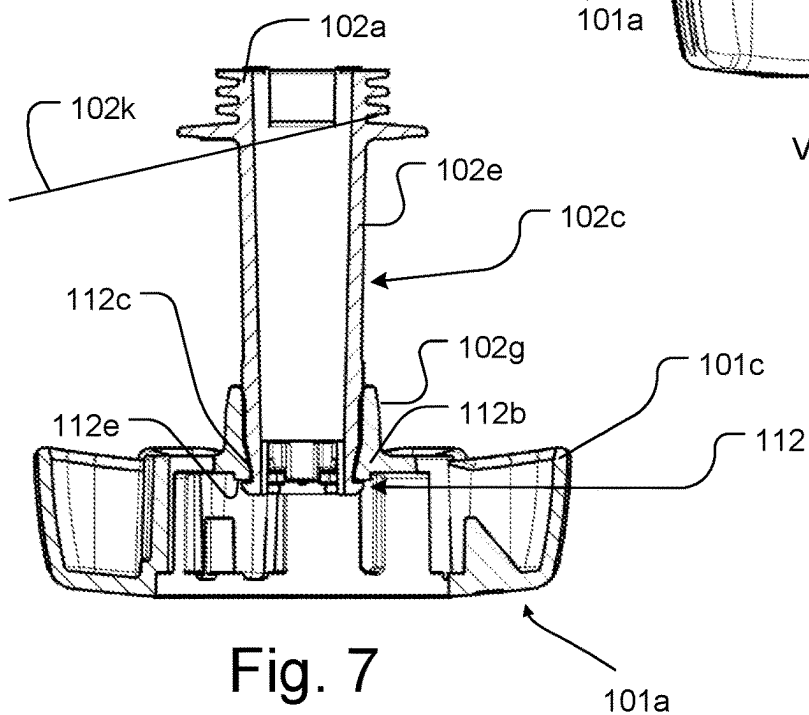
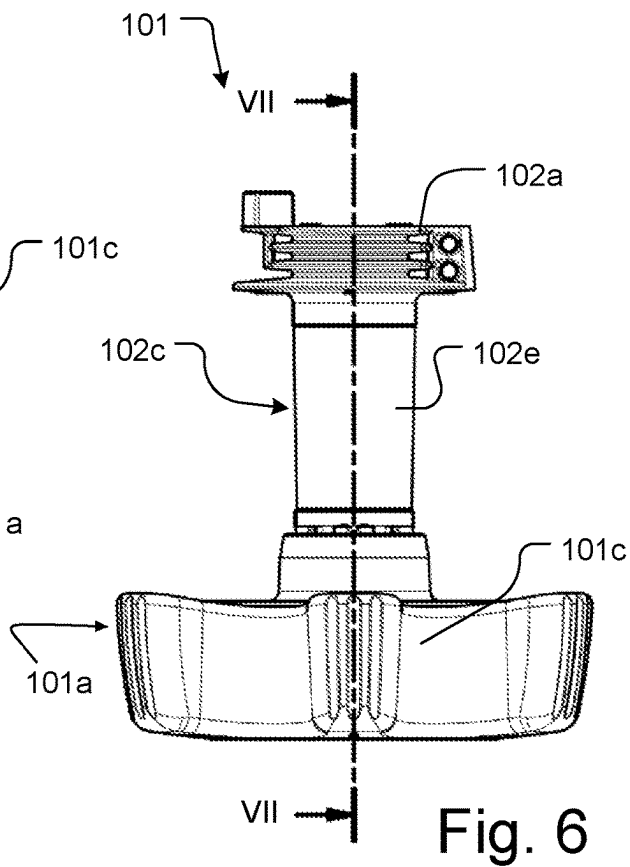
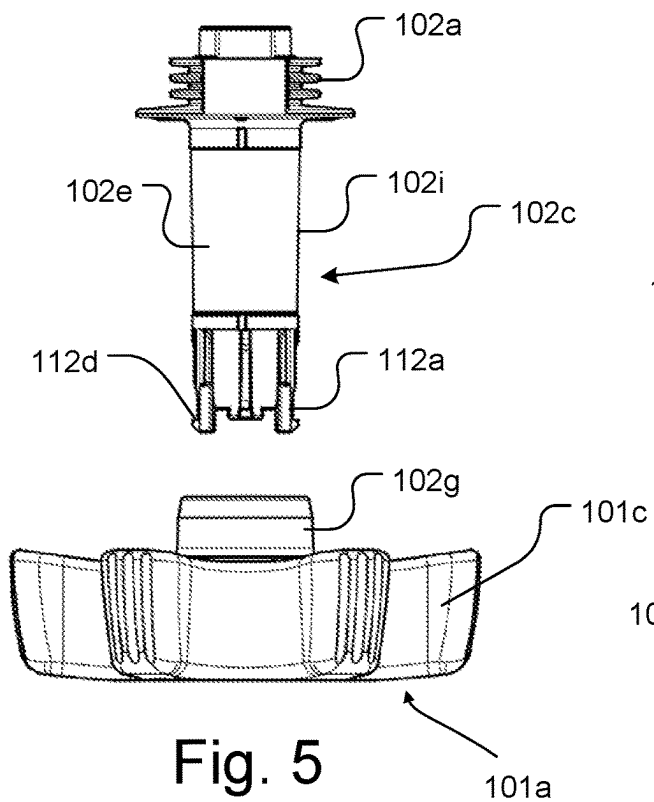
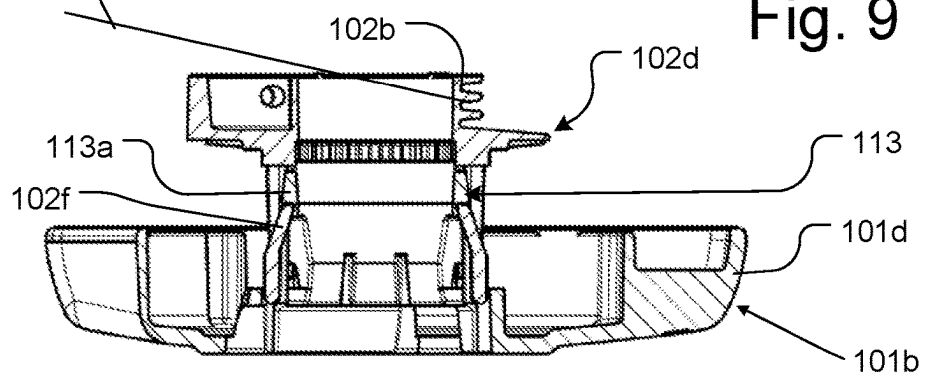
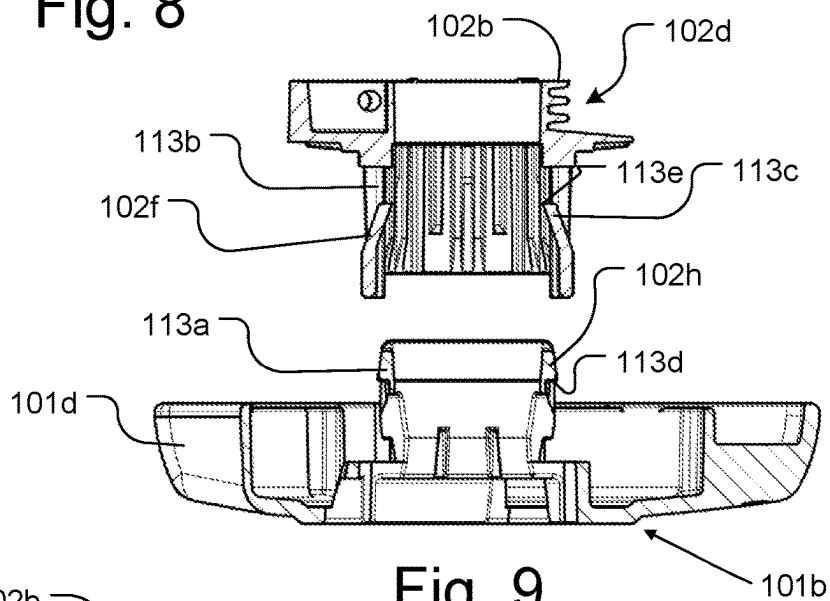
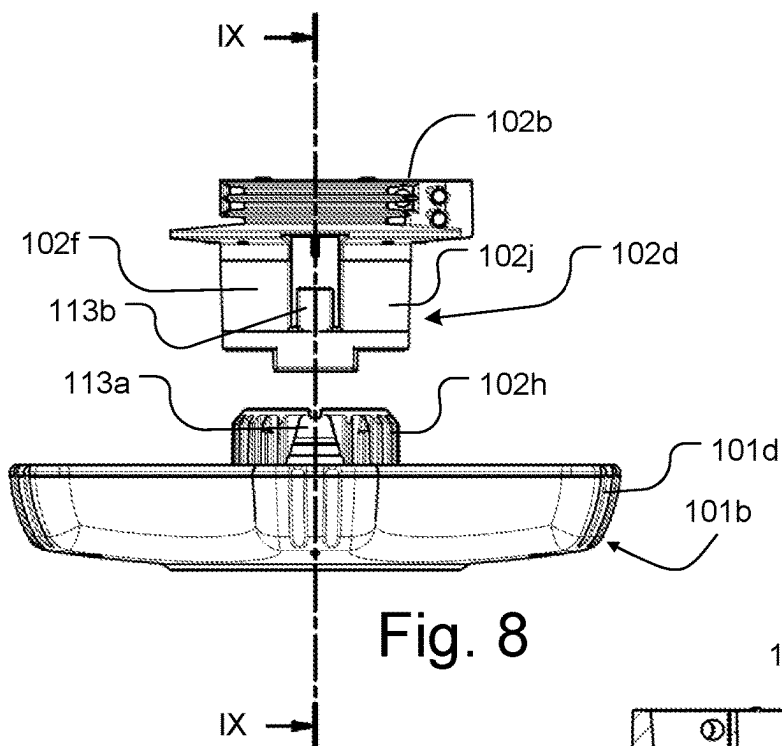


Fig. 4





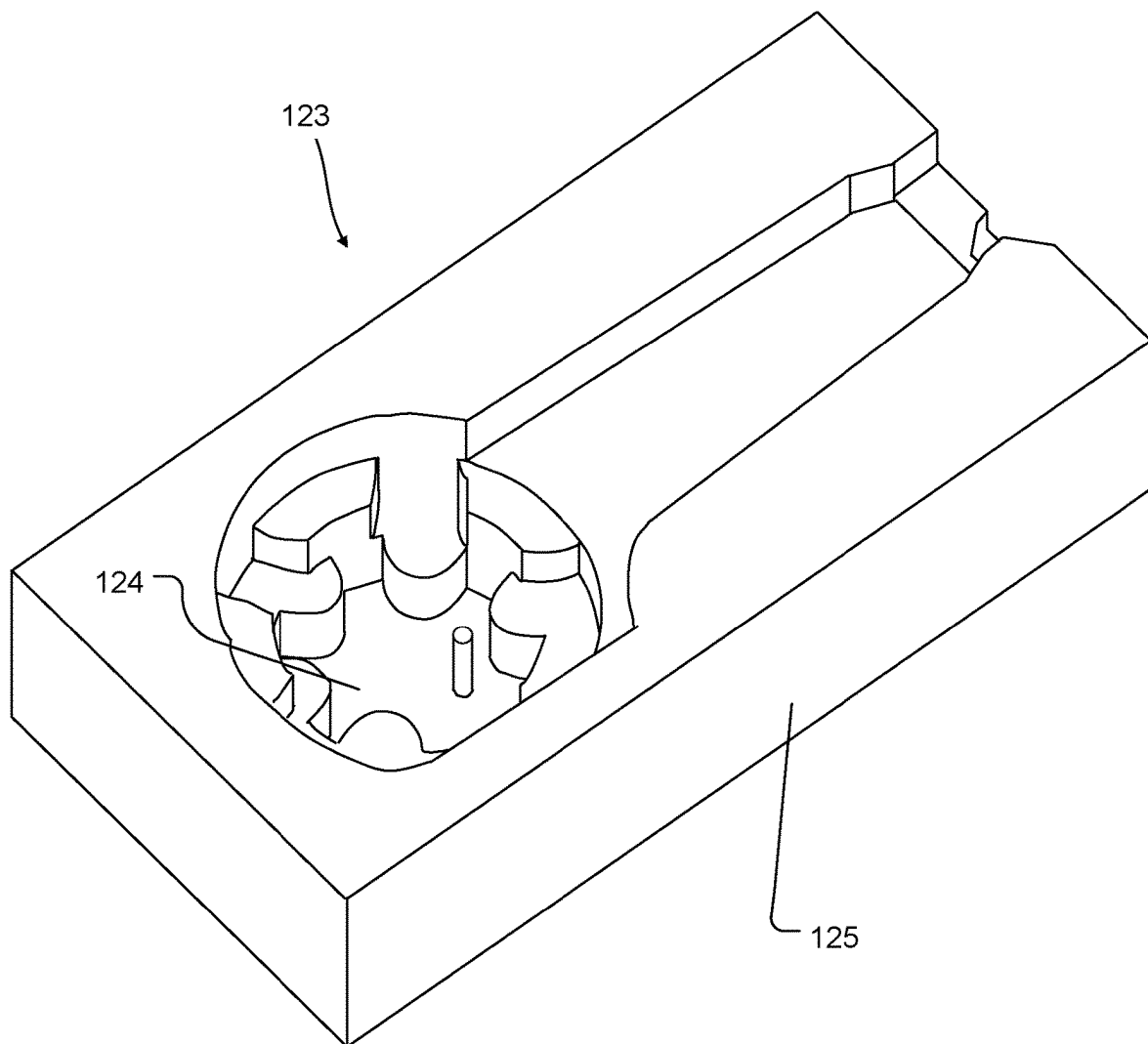


Fig. 11

1

METHOD OF ASSEMBLY OF AN ENDOSCOPE CONTROL SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a division of U.S. patent application Ser. No. 17/239,372, filed Feb. 24, 2023, which claims priority to and the benefit of European Patent Application Nos. 20172237.8 and 20172238.6, filed Apr. 30, 2020; the disclosures of said applications are incorporated by reference herein in their entirety.

TECHNICAL FIELD

The present disclosure relates to insertable medical vision devices, such as, but not limited to, endoscopes, in particular disposable insertion endoscopes, such as duodenoscopes, gastroscopes, and colonoscopes, and to methods of assembly thereof. More specifically, the present disclosure relates to endoscope control systems comprising control wheels connected to associated wire drums for connection to steering wires, whereby rotation of the control wheels controls a bending operation of a tip of the endoscope, and to assembly of such control systems.

BACKGROUND

Endoscopes are typically equipped with a light source and a vision receptor including a vision or image sensor. Provided that enough light is present, it is possible for the operator to see where the endoscope is steered and to set the target of interest once the tip has been advanced thereto.

Endoscopes typically comprise an elongated insertion tube with a handle at the proximal end, as seen from the operator, and visual inspection means, such as a built-in camera including a vision sensor, at a distal end of the elongated insertion tube. This definition of the terms distal and proximal, i.e. "proximal" being the end closest to the operator and "distal" being the end remote from the operator, as used herein for endoscopes in general, is adhered to in the present specification. Electrical wiring for the camera and other electronics, such as one or more LEDs accommodated in the tip part at the distal end, runs along the inside of the elongated insertion tube from the handle to the tip part. A working or suction channel may run along the inside of the insertion tube from the handle to the tip part, e.g. allowing liquid to be removed from the body cavity or allowing for insertion of surgical instruments or the like into the body cavity. The suction channel may be connected to a suction connector, typically positioned at a handle at the proximal end of the insertion tube.

To be able to maneuver the endoscope inside the body cavity, the distal end of some endoscopes comprises a bendable distal tip, which may be bendable in one, e.g. an up/down dimension, or two dimensions, e.g. an up/down dimension and a left/right dimension. The bendable tip often comprises a bending section with increased flexibility, e.g. achieved by articulated segments of the bending section. The maneuvering of the endoscope inside the body is typically done by tensioning or slacking steering wires also running along the inside of the elongated insertion tube from the tip part through the remainder of articulated segments to a control system or control mechanism positioned in or forming part of the handle.

An endoscope control system for performing a bending operation in two dimensions is known from

2

WO2018022418A2. This control system includes two control wheels connected to associated wire drums for connection to associated steering wires of the endoscope, whereby rotation of the control wheels controls the bending operation in two dimensions.

U.S. Pat. No. 4,461,282 discloses another endoscope control system including two control wheels.

U.S. Pat. No. 5,329,887 discloses another endoscope control system including two control wheels and in which a separately provided snap ring or snap clip attaches a control shaft.

In the prior art, other separate attachment means, such as locking rings or spring clips, have also been applied to attach parts to each other.

Furthermore, the control systems of the prior art are often assembled in a manner which requires flipping the parts around the already assembled parts during the method of assembly.

SUMMARY OF EMBODIMENTS OF THE DISCLOSURE

A first aspect of the present disclosure relates to a method of assembly of an endoscope control system, the endoscope control system being for performing a bending operation in a disposable insertion endoscope,

wherein the endoscope control system comprises:

a housing frame for forming or for forming part of an endoscope handle housing, the housing frame comprising a connection hole;

a first control wheel unit comprising a first wheel handle; a second control wheel unit comprising a second wheel handle;

a first shaft unit, the first shaft unit comprising a first wire drum and a first shaft, the first shaft connecting the first control wheel unit to the first wire drum, the first wire drum being for connection to a first steering wire of the endoscope, whereby rotation of the first wheel handle relative to the housing frame about an axis of rotation controls the bending operation in a first dimension;

a second shaft unit, the second shaft unit comprising a second wire drum and a second shaft, the second shaft connecting the second control wheel unit to the second wire drum, the second wire drum being for connection to a second steering wire of the endoscope, whereby rotation of the second wheel handle relative to the housing frame about an axis of rotation controls the bending operation in a second dimension;

wherein the method of assembly comprises, in sequence, the steps of:

1) holding the first control wheel unit in a position;
2) moving the second control wheel unit in an assembly direction to position the second control wheel unit on the first control wheel unit;

3) moving the housing frame in the assembly direction to position the housing frame on the second control wheel unit;

4) moving the second shaft unit in the assembly direction so that the second shaft is positioned to extend through the connection hole of the housing frame and snaps into engagement with the second control wheel unit by means of a second snap connection between the second shaft and the second control wheel unit; and

5) moving the first shaft unit in the assembly direction so that the first shaft is positioned to extend through the connection hole of the housing frame and through the second shaft and snaps into engagement with the first

3

control wheel unit by means of a first snap connection between the first shaft and the first control wheel unit; whereby the first and second control wheel units, the housing frame, and the first and second shaft units are maintained in position relative to each other in the assembly direction by means of the first and second snap connections.

The methods of the present disclosure may allow for mounting and/or positioning all the parts involved, i.e. the first and second control wheel units, the housing frame, and the first and second shaft units from one side only and in one assembly direction only, and/or wherein, in each of the steps 2) to 5), only the next part to be added is moved in one direction only.

The first and second snap connections may ensure that the parts of the control system assembled in steps 1) to 5), and potentially any further steps during or in between steps 1) to 5), are attached to each other only by means of the movement carried out in steps 4) and 5). Accordingly, the movement in each of steps 4) and 5) may activate the snap connections, potentially without any further action being required to activate these.

Furthermore, locking rings or similar separate locking elements as used in the prior art for attaching the first and second control wheel units, the housing frame, and the first and second shaft units can be replaced by the first and second snap connections so that separate locking elements can entirely be avoided in the attachment of these parts to each other. Accordingly, the methods according to the present disclosure may involve that no locking rings and/or separate locking elements are applied during and/or between steps 1) to 5) of the method.

Furthermore, when the parts are assembled in the sequence of steps 1) to 5) in one single assembly direction, flipping around or turning the already assembled parts may be avoided during the sequence of steps 1) to 5).

If a jig as described below is used during the method steps 1) to 5), it is possible to avoid removing the already assembled parts from the jig during steps 1) to 5) and any further steps carried out simultaneously with or in between steps 1) to 5).

Thus, simpler and/or faster assembly of the control system may be achieved with the methods of assembly and control systems according to the present disclosure.

Furthermore, a lower error rate may be achieved, and one or more of or all the method steps can be carried out on a machine, which may further increase the above advantages.

The housing frame may be a first half shell of the handle housing, the handle housing potentially further comprising a second half shell that may be attached to the first half shell after completion of steps 1) to 5). Hereby, the first and second wire drums may be positioned inside the assembled handle housing.

The first and/or second wheel handles may be generally circular and/or comprise finger depressions.

Each of the first and second control wheel units may comprise a central part or wheel sleeve, first and second, respectively, each potentially surrounding a center opening. The central parts may be cylindrical and may extend towards the housing frame in the assembled control system. In the assembled control system, the first central part may extend into the second central part, which may help proper positioning of the control wheel units in relation to each other during assembly. The second central part may extend to encompass part of the first central part in the assembled state of the control system.

4

The term "cylindrical" as used herein may involve conical or frusto-conical shapes.

The first shaft may comprise a bearing surface, and the second shaft may comprise a bearing surface, these bearing surfaces being for abutment and rotational sliding on associated bearing elements potentially fixed to or in one piece with the housing frame, see further below.

The first and/or second shafts may each be tubular and/or comprise a substantially cylindrical circumferential wall which may provide the associated bearing surfaces. A cross section or diameter of the first shaft may be smaller than that of the second shaft.

The axes of rotation may be coinciding. The first and second shafts, and potentially the first and second control wheel units and/or the first and second wheel handles, may extend coaxially in the assembled control system.

In the assembled control system, the first and second wire drums on one hand and the first and second wheel handles on the other hand may be positioned on opposite sides of the housing frame. The first wire drum may be positioned in extension of the second wire drum and/or farther away from the housing frame than the second wire drum.

The first and second wire drums may be positioned at ends of the first and second shafts, respectively.

The assembled control system may further include an inner bearing element which in a further step before, during, or after step 4) may be moved in the assembly direction to extend through the second shaft unit and/or through the connection hole of the housing frame. In the assembled control system, the inner bearing element may be rotationally fixed to the housing frame. The inner bearing element may comprise an inner bearing surface which may be positioned farther from a common axis of rotation than the outer bearing surface of the first shaft, the inner bearing surface abutting the outer bearing surface so that rotation of the second control wheel unit is at least partly borne on the inner bearing element. The inner bearing element may include a cylindrical and/or tubular inner bearing sleeve having a wall. The inner bearing element may separate rotation of the control wheel units from each other.

The assembled control system may further comprise an outer bearing element which may be in one piece with or fixed to the housing frame. The outer bearing element may surround and/or define the connection hole of the housing frame. The outer bearing element may be tubular and/or cylindrical and may comprise a wall which may extend away from the housing frame in a direction towards the first control wheel unit. The outer bearing element may comprise an inner bearing surface which may be positioned farther from a common axis of rotation than the outer bearing surface of the second shaft, the inner bearing surface abutting the outer bearing surface so that rotation of the second control wheel unit is at least partly borne on the outer bearing element. The outer bearing element may be an outer bearing sleeve that encompasses at least part of the second shaft.

The control system may further include a center shaft which in step 5) or in a further step before or, in particular, after step 5) may be moved in the assembly direction to extend through the connection hole of the housing frame, the first and second shaft units, and the center openings of the first and second control wheel units. The center shaft may comprise or may be attached or fixed to a connector frame, and the connector frame may be attached or fixed to the housing frame after insertion of the center shaft so that the center shaft is fixed to the housing frame. This may provide improved rigidity and structural stability of the assembled

5

control system. The connector frame can be attached or fixed to the housing by one or more snap connections, screws, connector pins, or any other suitable means.

The center shaft may include a center shaft arm or a center shaft frame that extends radially from a shaft part thereof and is to be positioned within the handle housing in the assembled endoscope. Similarly, the inner bearing element may include an inner bearing element arm or an inner bearing element frame that extends radially from a sleeve part thereof and is to be positioned within the handle housing in the assembled endoscope. Each of these arms or frames may be fixed to the housing frame in order to attach or fix the center shaft and/or inner bearing element to the housing frame. This attachment or fixation may be achieved by one or more snap connections, screws, connector pins, or any other suitable means. The inner bearing element arm or frame may be directly attached or fixed to the housing frame, the center shaft arm or frame being directly attached or fixed to the inner bearing element arm or frame so as to be indirectly attached or fixed to the housing frame, in which case the inner bearing element arm or frame may be positioned between the center shaft arm or frame and the housing frame.

When the center shaft has been positioned, a cap may be moved opposite to the assembly direction and be attached at and/or to a tip end of the center shaft. This attachment may involve a further snap connection which may be provided in a manner identical or similar to the first and second snap connections as described herein. Accordingly, the cap may include one or more, such as two, resilient/pushable connection parts, whereas the tip end of the center shaft may include associated one or more connection parts taking the form of, for example, recesses. This snap engagement may similarly be activated during or at the end of the insertion of the center shaft into the control system. The cap may thus, be positioned and held in the jig described below before or at the same time as the first control wheel unit is positioned in the jig. The cap may cover and/or attach a first brake of the first control wheel unit, see further below regarding the brake. The brake may be encased or housed within a spacing defined by interior surfaces of the first wheel handle and the cap. If the first control wheel handle includes such a brake, it may be advantageous to, instead, remove control system from the jig after steps 1) to 5), then potentially flipping the assembled control system around, e.g. a 180 degrees rotation, after which the first brake is assembled and positioned to form part of the first control wheel unit, after which, again, the cap is positioned as described. Alternatively, the first brake and the cap are mounted to form part of the first control wheel unit before or during step 1). The cap may also be included even if no brake is included in the first control wheel unit; in this case, the cap and associated snap connection may simply be used to attach or hold the tip part of the center shaft.

In the assembled control system, a brake handle for activation of a second brake, which is the brake of the second control wheel unit described further below, may be attached to the housing frame. During the method of assembly, when the second control wheel unit has been positioned in step 2), before step 3), the brake handle may be moved in the assembly direction to be positioned on the second control wheel unit. In step 3), the housing frame may then be attached to the brake handle. This attachment may again, involve a further snap connection which may be provided in a manner identical or similar to the first and second snap connections as described herein. Accordingly, the housing frame, such as the outer bearing element of the housing

6

frame, may include one or more, such as two, resilient/pushable connection parts, whereas the brake handle includes associated one or more connection parts taking the form of, for example, projections. This snap engagement may similarly be activated during or at the end of the positioning movement of the housing frame.

In step 1), the first control wheel unit may be positioned in a jig, the jig holding the first control wheel unit, potentially in a fixed position. A jig may be any device suitable for holding at least the first control wheel unit in position during the assembly process.

In step 2), the second control wheel unit is also positioned and held in the jig.

In step 3), the housing frame is also positioned and held in the jig.

The jig may include a jig block with a jig depression shaped to provide a positive engagement with the first control wheel unit, especially the first wheel handle. Similarly, the jig depression may be shaped to also provide a positive engagement with the second control wheel unit, especially the second wheel handle, when the second control wheel unit is positioned on the first control wheel unit in the jig. Similarly, the jig depression may be shaped to also provide a positive engagement with the housing frame or a part thereof when the housing frame is positioned on the second control wheel unit in the jig. Hereby, all of these elements can be maintained in relative positions until the snap connections are engaged in steps 4) and 5).

After steps 1) to 5), the assembled control system may be removed from the jig, and further assembly of the endoscope may occur.

In step 2), the second control wheel unit may be positioned in abutment with the first control wheel unit, or they may be positioned at a distance from each other, a further element potentially providing this distance, or the distance being achieved during the subsequent steps of assembly.

In step 3), the housing frame may be positioned in abutment with the second control wheel unit, or they may be positioned at a distance from each other, a further element potentially providing this distance, or the distance being achieved during the subsequent steps of assembly.

One or both snap connections may be snap fittings.

The second snap connection may axially and/or rotationally fix the second shaft unit to the second control wheel unit. The second snap connection may include one or more, such as two, connection parts of the second shaft unit interlocking with one or more, such as two, associated connection parts of the second control wheel unit. These connection parts may be any suitable mutually engaging snap connection parts. During the movement of the second shaft unit, the one or more connection parts of one of the second shaft unit and the second control wheel may be pushed in a radial direction (inwardly or outwardly) and, then, when the second shaft unit is further moved or inserted, resiliently snap back to engage the associated one or more connection parts. As is conventional, this may be achieved by the one or more of the connection parts that are not pushed including a ramp or an inclined surface, which forces the pushable connection parts in the radial direction during the movement of the second shaft unit. Accordingly, one or more of the pushable connection parts may include a barb surface which during the snap moves into engagement with an associated barb surface of the other connection parts to secure the position of the second shaft unit to the second control wheel unit. For example, the pushable connection parts may be included in the second control wheel unit, potentially in the central part thereof, and the associated

7

connection parts may be included, potentially as simple holes or recesses, in the second shaft unit. The pushable connection part(s) may be provided by two axially extending slots defining a resilient, axially moveable pin between them.

Like the second snap connection, the first snap connection may axially and/or rotationally fix the first shaft unit to the first control wheel unit. And the first snap connection may include one or more, such as two, connection parts of the first shaft unit interlocking with one or more, such as two, associated connection parts of the first control wheel unit. These connection parts may be any suitable mutually engaging snap connection parts. During the movement of the first shaft unit, the one or more connection parts of one of the first shaft unit and the first control wheel may be pushed in a radial direction (inwardly or outwardly) and, then, when the first shaft unit is further moved or inserted, resiliently snap back to engage the associated one or more connection parts. And again, this may be achieved by the one or more of the connection parts that are not pushed including a ramp or an inclined surface, which forces the pushable connection parts in the radial direction during the movement of the first shaft unit. Accordingly, one or more of the pushable connection parts may include a barb surface which during the snap moves into engagement with an associated barb surface of the other connection parts to secure the position of the first shaft unit to the first control wheel unit. For example, the pushable connection parts may be included in the first shaft unit, potentially in the first shaft thereof, and the associated connection parts may be included, potentially as simply holes or recesses, in the first control wheel unit. The pushable connection part(s) may be provided by two axially extending slots defining a resilient, axially moveable pin between them.

Hereby, the first and second snap connections, when engaged, prevent movement of the first and second control wheels, the housing frame, and the first and second shaft units relative to each other in the assembly direction, which may be an axial direction of the control system. Accordingly, the barb surfaces may face in the axial direction. The barbs may be provided as projections extending in a radial direction.

The assembly direction may be parallel with the axes of rotation of the first and second control wheel units.

The housing frame may include one or more screw towers for fastening of the screws described above. The screw towers may extend oppositely to the assembly direction.

In the assembled control system, one of or both the first and second control wheel units may comprise a brake. Activation of the brake may move the brake from a released position to a braking position. A brake force of the brake in the braking position may brake rotation of the associated control wheel unit. The brake force may in the released position be at least partially released. The brake may be a multi-disc brake, which may include a stack of at least three brake discs. Each brake may include a handle or a button, which may activate the brake, e.g. by rotation of the handle or button. Each brake may comprise a spring, which may be a helical spring. Each brake may be assembled in or on the associated control wheel unit or associated wheel handle before or after or during the associated steps of positioning the first and second control wheel units, i.e. steps 1) and 2), respectively. A second such brake may be assembled in the second control wheel unit or the second wheel handle before step 1) or before step 2). A first such brake may be assembled in the first control wheel unit or the first wheel handle after step 4) or after step 5). In case a jig is applied as described

8

above, the method of assembly may include the further step of removing the control system from the jig after step 5). After the step of removing the control system from the jig, the method of assembly may include the further step of assembling the first brake in the first control wheel unit.

The endoscope control system can alternatively be denoted an endoscope bending operation apparatus.

The control system may be positioned on or in, or may form part, of an endoscope handle of the endoscope, see also further below.

The first and/or second control wheel units and/or the inner and/or outer bearing element, and/or the first and/or second shaft units may form part of the endoscope handle for or of an endoscope.

The first control wheel unit may be positioned coaxially with the second control wheel unit and/or with the outer bearing member and/or the inner bearing member, and/or the first and/or second shaft units.

The handle housing may be manufactured of a plastic material, potentially a plastic polymer material and/or an artificial resin. One or both control wheels or at least the sleeves thereof described below, and/or one or both wire drums may be manufactured from one or more plastics or a plastics material, such as POM. The entire control system may be manufactured of one or more plastics or plastic materials or plastic polymer materials or artificial resins, and/or the control system does not include any metal. Any one of the plastic materials mentioned herein may be a plastic polymer material which comprises or consists of one or more of PC, polypropylene (PP), acrylonitrile butadiene styrene (ABS), polyethylene (PE), polyamide (PA), polyurethane (PU), polystyrene (PS), polylactic acid (PLA), polyvinyl chloride (PVC), polyoxymethylene (POM), polyester, polyethylene terephthalate (PET), and acrylic (PMMA). The polymer may be a copolymer of one or more monomers of the latter materials.

Rotation of the control wheel units may occur relative to the housing frame. The handle housing may be a handle shell or a housing shell. The housing frame and/or the handle housing may be manufactured of a rigid material, such as a rigid plastic polymer.

The first and/or second wire drums may be a pulley/pulleys. The first and second steering wires may form part of the control system and may be attached to the first and the second drum, respectively, and/or may be wound up or woundable on these, respectively.

The axis of rotation may be an axis of rotation of also the first and/or second shafts.

The axis of rotation may be a center axis of the first and/or second control wheel units and/or of the first and/or second wheel handles and/or of the first and/or second shaft units and/or shafts and/or of the control system and/or of the outer bearing element and/or of the inner bearing element and/or of the center shaft.

The second wheel handle may be positioned between the handle housing and the first wheel handle. The first shaft unit may extend through the second wheel handle and/or through the second shaft unit, and/or through the second wire drum. The first shaft unit may extend through center holes or center openings of these elements.

The second wire drum may be positioned between the first wire drum and the second control wheel unit.

The first control wheel unit may be an inner control wheel and/or may be positioned closer to the housing frame than the second control wheel unit, and/or may control a bendable tip of the endoscope in an up/down dimension. The second

control wheel unit may be an outer control wheel and/or may control a bendable tip of the endoscope in a left/right dimension.

The second shaft may encompass at least part of the first shaft. The first and/or second shafts may each be cylindrical or circular cylindrical and/or may be embodied as cylinder shells and/or may be hollow. The second shaft may be positioned to be coaxial with the first shaft. The inner bearing element may be an inner bearing sleeve and may be the only element positioned between the first shaft and the second shaft.

Any one of or all of the above-mentioned elements of the control system, potentially except for steering wires and/or parts of the potential brakes may be manufactured from plastic polymer(s).

The second control wheel unit may be positioned coaxially with and potentially axially shifted in relation to the first control wheel unit. A diameter or a cross-sectional dimension of the two wheel handles may be different from each other, potentially so that an outer one of the two wheel handles has a smaller diameter or smaller cross-sectional dimension.

The outer bearing element may be in one piece with the housing frame and/or molded in one piece with the handle housing. Alternatively, the outer bearing element is provided separately from and is fixed to the handle housing.

Any one or more of or all steps 1) to 5) may be carried out automatically. Any one or more or all of potential further steps of the method as disclosed above may also be carried out automatically.

A step being carried out automatically may involve that the step is carried out by means of a machine, wherein the machine may be or comprise a robot, which may comprise a robotic arm and a control system. The jig may form part of the machine. The jig may be replaced by a holding or gripping mechanism or hand of a robotic arm.

Alternatively or additionally, any one or more or all of the method steps of the methods according to the present disclosure may be carried out manually, such as a person using human hands and/or one or more tools for carrying out the step in question.

A combination of automatic and manual assembly may also be envisioned, wherein any one or more of the steps are carried out automatically as described and the remaining steps are carried out manually as described.

In an embodiment of the methods of assembly according to the present disclosure, all the movements of steps 2) to 5) are carried out from one side only.

Hereby, a need to flip around the already assembled parts during the method of assembly may be avoided.

In another embodiment, in each of steps 2) to 5), all already positioned parts of the control system to be assembled remain in a held position.

In another embodiment, only the movement carried out in steps 4) and 5) affect the attachment of the parts assembled in steps 1) to 5) to each other.

In other words, the first and second snap connections may be the only connections affecting the attachment of the parts to each other. The snap connections may generally activate automatically during and because of the movement in steps 4) and 5).

It is understood that connection parts of the snap connections may also move in themselves, typically in a radial direction, during the snapping action, as described in the above.

In another embodiment, during the sequence of steps 1) to 5), no separate locking device is applied to attach the parts to each other.

In another embodiment, the first and second snap connections are provided only by snap connection parts provided in one piece with the first shaft unit, the first control wheel unit, the second shaft unit, and the second control wheel unit, respectively.

In another embodiment, the first snap connection comprises at least one primary connector part forming part of the first shaft unit and at least one secondary connector part forming part of the first control wheel unit, the primary and secondary connector parts in step 5) snapping directly onto each other to form the first snap connection; and the second snap connection comprises at least one tertiary connector part forming part of the second shaft unit and at least one quaternary connector part forming part of the second control wheel unit, the tertiary and quaternary connector parts in step 4) snapping directly onto each other to form the second snap connection.

Any one or more of the primary to quaternary connection parts may be provided as the pushable/moveable/resilient connection parts described above. The other one of the primary and secondary connection parts and the tertiary and quaternary connection parts, respectively, may then be the static/non-moveable connection parts.

Any one or more of or all the connection parts may include a ramp facilitating the resilient movement of one or more of the pushable connection parts during the steps 4) and/or 5).

In another embodiment, no further steps are involved in the method during the sequence of steps 1) to 5).

In another embodiment, before step 1), the first wheel is positioned in a jig, the first wheel being held in the jig in step 1) and during the sequence of steps 2) to 5).

The method may generally, further comprise the step of, after step 5), removing the parts assembled in steps 1) to 5) from the jig.

In another aspect, the present disclosure involves an endoscope control system for performing a bending operation in a disposable insertion endoscope, wherein the endoscope control system comprises:

- a housing frame for forming or for forming part of an endoscope handle housing, the housing frame comprising a connection hole;

- a first control wheel unit comprising a first wheel handle;
- a second control wheel unit comprising a second wheel handle;

- a second shaft unit, the second shaft unit comprising a second wire drum and a second shaft, the second shaft connecting the second control wheel unit to the second wire drum, the second wire drum being for connection to a second steering wire of the endoscope, whereby rotation of the second wheel handle relative to the housing frame about an axis of rotation controls the bending operation in a second dimension;

wherein the second control wheel is positioned between the first control wheel and the housing frame, the second shaft extends through the connection hole of the housing frame, and the first shaft extends through the connection hole of the housing frame and through the second shaft; and

wherein the first shaft is connected to the first control wheel unit by means of a first snap connection between the first shaft and the first control wheel unit, and the second shaft is connected to the second control wheel

11

unit by a second snap connection between the second shaft and the second control wheel unit, whereby the first and second snap connections attach the first and second control wheels, the housing frame, and the first and second shaft units to each other.

The control system according to the present aspect may include any one or more of the embodiments and/or structural features disclosed above in relation to the control systems assembled according to the methods of the present disclosure.

In an embodiment of the control system according to the present aspect, the first and second snap connections maintain attachment between the first and second control wheels, the housing frame, and the first and second shaft units if all other parts are removed.

In another embodiment, the first and second snap connections are provided only by snap connection parts provided in one piece with the first shaft unit, the first control wheel unit, the second shaft unit, and the second control wheel unit, respectively.

In another embodiment, the endoscope control system has been assembled according to the method of any one of the above embodiments of methods of assembly.

In another aspect, the present disclosure involves an endoscope comprising the control system assembled according to any one or more of the above embodiments of methods of assembly and/or comprising the control system according to any one or more of the above embodiments.

In an embodiment of the endoscope according to the present aspect, the endoscope further comprises the first and second steering wires and a distal tip or tip part that comprises a bending section connected to the first and second steering wires so that the control system can activate the bending operation of the bending section via the steering wires.

The endoscope may further comprise an endoscope handle at the proximal end thereof, and/or visual inspection means, such as a built-in camera including a vision sensor, at a distal tip. Electrical wiring for the camera and other electronics, such as one or more LEDs accommodated in the tip part at the distal end, may run along the inside of the elongated insertion tube from the endoscope handle to a PCB or an FPC at the distal tip. A working or suction channel may run along the inside of the insertion tube from the handle to the tip part, e.g. allowing liquid to be removed from the body cavity or allowing for insertion of surgical and/or sampling instruments or the like into the body cavity. The suction channel may be connected to a suction connector, typically positioned at a handle at the proximal end of the insertion tube.

In some embodiments of the endoscope, the endoscope further comprises a distal tip or tip part that comprises a bending section connected to the steering wire(s) so that the control system can activate a bending operation of the bending section via the steering wire(s).

The bending section may be bendable in one or two dimensions, e.g. an up/down dimension and a left/right dimension. The bendable tip may comprise a bending section with increased flexibility, e.g. achieved by articulated segments of the bending section as are known in the art. The steering wire(s) may run along the inside of an elongated insertion tube from the tip through the bending section to the control system positioned in or forming part of the endoscope handle.

The endoscope may be a disposable insertion endoscope. The endoscope may include one or more features as described herein in the above, including the features of

12

endoscopes described in the above introduction to this description, and in connection with the description of the methods and tip parts according to the present disclosure.

The endoscope may comprise an elongated insertion tube with a handle at the proximal end. A tip or tip part may be positioned at the distal end of the elongated insertion tube. The tip may further comprise a bending section positioned between the tip and the elongated insertion tube. The bending section may be configured to be articulated to maneuver the endoscope inside a body cavity.

The endoscope may be a duodenoscope, a gastroscope, or a colonoscope.

An embodiment of the endoscope further comprises the first and second steering wires and a distal tip or tip part that comprises a bending section connected to the first and second steering wires so that the control system can activate the bending operation of the bending section via the steering wires.

A person skilled in the art will appreciate that any one or more of the above aspects of this disclosure and embodiments thereof may be combined with any one or more of the other aspects and embodiments thereof.

BRIEF DESCRIPTION OF THE DRAWINGS

In the following, non-limiting exemplary embodiments will be described in greater detail with reference to the drawings, in which:

FIG. 1 shows a perspective view of an endoscope including a control system according to the present disclosure;

FIG. 2 shows a top view of the control system of FIG. 1;

FIG. 3 shows a cross-sectional view taken along line III-III in FIG. 2;

FIG. 4 shows an exploded perspective view of a handle frame and the control system of the endoscope of FIG. 1;

FIG. 5 shows an exploded side view of a first control wheel unit and a first shaft unit of the control system of FIG. 1;

FIG. 6 shows an exploded side view of the first control wheel unit and first shaft unit of FIG. 5 in an assembled state and in which the first control wheel unit and first shaft unit have been turned 180 degrees;

FIG. 7 shows a cross section taken along the line VII-VII in FIG. 6;

FIG. 8 shows an exploded side view of a second control wheel unit and a second shaft unit of the control system of FIG. 1;

FIG. 9 shows a cross-sectional view taken along the line IX-IX of FIG. 8;

FIG. 10 shows a view like that of FIG. 9, wherein the second control wheel unit and second shaft unit are in an assembled state; and

FIG. 11 shows a perspective view of a jig.

DETAILED DESCRIPTION

In this disclosure, the term “to accommodate” may additionally or alternatively be defined as “to house” or “to enclose” or “to surround”.

In this specification, the terms “integrally” or “integrally provided” or “integrally comprising”, “in one piece”, or similar may be defined as the associated features forming an integral part of a whole; and/or are in one piece, potentially molded in one piece; and/or are substantially inseparable by hand. When a first element forms part of a second element, this may involve that the first element is provided integrally with or in one piece with the second element.

13

As mentioned, in this specification, the term “proximal” may be defined as being closest to an operator of the endoscope, and the term “distal” as being remote from the operator. The term “proximal-distal” may be defined as extending between these two extremes, in the present case proximal-distal may extend along a center axis of the tip part extending between a proximal extremity of the proximal end of the tip part and a distal extremity of the distal end of the tip part.

In this specification, an endoscope may be defined as a device adapted for viewing body cavities and/or channels of a human and/or animal body. The endoscope may for instance be a flexible or steerable endoscope. The endoscope may be a duodenoscope or a ureteroscopy, a gastroscope, or a colonoscope.

FIG. 1 shows a disposable insertion endoscope 1 with a control system 100, an elongated insertion tube 3, and an endoscope handle 2 at a proximal end 3a of the elongated insertion tube 3. The endoscope handle includes a handle housing 116

In a known manner, an endoscope tip 4 is positioned at a distal end 3b of the elongated insertion tube 3, the tip 4 comprising a bending section 5 positioned between the tip 4 and the elongated insertion tube 3. The endoscope handle 2 comprises the endoscope control system 100, the endoscope control system 100 being for performing a bending operation of the disposable insertion endoscope 1.

In a known manner, the bending section 5 is connected to steering wires, which extend from the control system 100 through the tube 3 to allow the control system 100 to activate a two-dimensional bending operation of the bending section 5 via the steering wires. The bending section 5 is configured to be articulated to maneuver the endoscope 1 inside a body cavity (not shown). The bending section 5 is bendable in two dimensions, i.e. an up/down dimension and a left/right dimension. In an alternative, not shown embodiment, the bending section is bendable in one dimension only.

The bending section 5 has increased flexibility achieved by articulated segments of the bending section 5 as is known in the art. The steering wires run along the inside of the elongated insertion tube 3 from the tip 4 through the bending section 5 to the endoscope control system 100. Still in a known manner, the maneuvering of the endoscope 1 inside the body can be carried out by tensioning or slacking the steering wires by means of the control system 100.

Still in a known manner, the distal tip 4 has a not shown built-in camera including a vision sensor. Not shown electrical wiring for the camera and potential other electronics, such as one or more LEDs accommodated in the tip part 4, run along the inside of the elongated insertion tube 3 from the endoscope handle 2 to a PCB or an FPC at or in the distal tip 4. A not shown suction/working channel runs along the inside of the insertion tube 3 from the handle 2 to the tip part 4, e.g. allowing liquid to be removed from the body cavity or allowing for insertion of a surgical instrument and/or a sampling instrument or other instruments (not shown) into the body cavity. The suction channel is connected to a suction connector 6 positioned at the proximal end of the handle 2. A sampling connector 81 is positioned at the distal end of the handle 2.

The endoscope control system 100 is shown exploded in FIG. 4 and comprises a housing frame 116a forming part of the handle housing 116 in the assembled endoscope 1, the housing frame 116a forming a half part of the assembled handle housing 116. The housing frame 116a at one end comprises a connection hole 116b.

14

The control system 100 further comprises a first control wheel unit 101a comprising a first wheel handle 101c comprising finger depressions and a second control wheel unit 101b comprising a second wheel handle 101d similarly comprising finger depressions.

The control system 100 further comprises a first shaft unit 102c and a second shaft unit 102d. The first shaft unit 102c comprises a first wire drum 102a and a first, sleeve-shaped shaft 102e, the first shaft 102e connecting the first control wheel unit 101a to the first wire drum 102a, the first wire drum 102a being for connection to a first steering wire 102k (shown in FIG. 7) of the endoscope 1, whereby rotation of the first wheel handle 101c relative to the housing frame 116a about an axis of rotation controls the bending operation in a first dimension. The second shaft unit 102d comprises a second wire drum 102b and a second shaft 102f; the second shaft 102f connecting the second control wheel unit 101b to the second wire drum 102b, the second wire drum 102b being for connection to a second steering wire 102l (shown in FIG. 10) of the endoscope 1, whereby rotation of the second wheel handle 101d relative to the housing frame 116a about an axis of rotation controls the bending operation in a second dimension. As described below, the first control wheel unit 101a and the first shaft unit 102c are coupled together in what may be referred to as a first control wheel 101 (best seen in FIG. 6), while the second control wheel unit 101b and the second shaft unit 102d are coupled together in what may be referred to as a second control wheel 102 (best seen in FIG. 10).

Referring to FIGS. 4 to 10, an embodiment of the methods of assembly of the control system 100 according to the present disclosure comprises, in sequence, the steps of:

- 1) holding the first control wheel unit 101a in a fixed position;
- 2) moving the second control wheel unit 101b in an assembly direction D to position the second control wheel unit 101b on the first control wheel unit 101a;
- 3) moving the housing frame 116a in the assembly direction D to position the housing frame 116a on the second control wheel unit 101b;
- 4) moving the second shaft unit 102d in the assembly direction D so that the second shaft is positioned to extend through the connection hole 116b of the housing frame 116a and snaps into engagement with the second control wheel unit 101b by means of a second snap connection 113 (shown in FIGS. 8-10) between the second shaft 102f and the second control wheel unit 101b; and
- 5) moving the first shaft unit 102c in the assembly direction D so that the first shaft 102e is positioned to extend through the connection hole 116b of the housing frame 116a and through the second shaft 102f and snaps into engagement with the first control wheel unit 101a by means of a first snap connection 112 (shown in FIGS. 5-7) between the first shaft 102e and the first control wheel unit 101a;

whereby the first and second control wheel units 101a, 101b, the housing frame 116a, and the first and second shaft units 102c, 102d are maintained in position relative to each other in the assembly direction D by means of the first and second snap connections 112, 113.

This method allows for mounting and positioning all the parts involved, i.e. the first and second control wheel units 101a, 101b, the housing frame 116a, and the first and second shaft units 102c, 102d from one side only and in the assembly direction D only. In each of the steps 2) to 5), only the next part to be added is moved in the assembly direction

15

while the already assembled parts are not moved. Mounting and positioning all the parts involved from one side only simplifies assembly and such simplification and corresponding manufacturing cost reduction is made possible by the present embodiment. As described below, a jig could be used to assemble the parts in sequence. While the method may be performed with the steps 1) to 5) performed in sequence, the first and second control wheel units **101a**, **102b** could be mated before placing them onto the jig, and the first and second shaft units **102c**, **102d** could be assembled together before snapping the shafts onto the first and second control wheel units **101a**, **102b**. In another example, the first and second shaft units **102c**, **102d** and the housing frame **116a** could be assembled together before snapping the shafts onto the first and second control wheel units **101a**, **102b**. In a further example, a jig could be used to assemble the system from the opposite direction, mounting the shafts on the jig and then, in the direction opposite D, snapping the first and second control wheel units **101a**, **102b** onto the first and second shaft units **102c**, **102d**. As can be deduced from the foregoing, the assembly advantages are derived from the structures of the components being assembled. The assembly direction D may be referred to as the axial direction.

The first control wheel unit, the second control wheel unit, the housing frame, the second shaft unit and the first shaft unit can be considered components of a stack, and the method can thus comprise stacking the components in order to form the stack. Stacking the components in order comprises holding a portion of the stack while moving the components onto the portion of the stack being held. The first control wheel unit can be considered part of the stack and can be held in the jig, as described, as the other components are stacked via movement in the axial direction toward the first control wheel unit. The stack may comprise an inner bearing element frame and a center shaft. The method may include, after moving the second shaft unit, moving the inner bearing element frame in the axial direction to position the inner bearing element frame through the connection hole of the housing frame. The inner bearing element frame has a portion positioned between the first shaft and the second shaft. The method may include, after moving the first shaft unit, moving a center shaft in the axial direction to position the center shaft through the first shaft and moving a cap in the axial direction toward the first control wheel unit to form a snap connection between a tip end of the center shaft and the cap to secure the center shaft in place.

The first and second snap connections **112**, **113** ensure that the parts of the control system assembled in steps 1) to 5) are attached to each other only by means of the movement carried out in steps 4) and 5). Accordingly, the movement in each of steps 4) and 5) activate the snap connections **112**, **113** without any further action being required to activate these. No locking rings or separate locking elements are applied during or between steps 1) to 5). Flipping around or turning the already assembled parts is avoided during the sequence of steps 1) to 5). A jig **123** as shown in FIG. **11** is used as described below during the method steps 1) to 5), and the already assembled parts do not need to be removed from the jig **123** during steps 1) to 5).

The housing frame **116a** is a first half shell of the handle housing **116a**, the handle housing **116a** further comprising a second half shell that is attached to the first half shell **116a** after completion of steps 1) to 5). Hereby, the first and second wire drums **102a**, **102b** are positioned inside the assembled handle housing **116**.

16

The first and second wheel handles **101c**, **101d** are generally circular and/or comprise conventionally provided finger depressions or cut-outs.

FIGS. **5-7** describe the components of and the first snap connection **112**, and FIGS. **8-10** describe the components of and the second snap connection **113**. Each of the first and second control wheel units **101a**, **101b** comprises a central part or wheel sleeve **102g**, **102h**, first and second, respectively, each surrounding a center opening. The central parts **102g**, **102h** are cylindrical and extend towards the housing frame **106a** in the assembled control system **100**. In the assembled control system **100**, the second central part **102h** may in other embodiments extend to encompass part of the first central part **102g** in the assembled state of the control system **100**.

The first shaft **102e** comprises a bearing surface **102i**, and the second shaft **102f** comprises a bearing surface **102j**, the bearing surfaces **102i**, **102j** being for abutment and rotational sliding on associated bearing elements fixed to the housing frame **116a**, see further below.

The first and second shafts **102e**, **102f** are each tubular and each comprises a substantially cylindrical circumferential wall which provide the associated bearing surfaces **102i**, **102j**. A diameter of the first shaft **102e** is smaller than that of the second shaft **102f**.

The axes of rotation of the control wheel units **101a**, **101b** are coinciding to form one axis of rotation, which is also a center axis of the control system **100**. This axis extends in the assembly direction D. The first and second shafts **102e**, **102f**, the first and second control wheel units **101a**, **101b**, and the first and second wheel handles **101c**, **101d** extend coaxially in the assembled control system **100**.

In the assembled control system **100**, as shown in FIG. **4**, the first and second wire drums **102a**, **102b** on the one hand and the first and second wheel handles **101c**, **101d** on the other hand are positioned on opposite sides of the connection hole **116b** of the housing frame **116a**. The first wire drum **102a** is positioned in extension of the second wire drum **102b** and farther from the housing frame **116a** or the connection hole **116b** than the second wire drum **102b**.

The first and second wire drums **102a**, **102b** are positioned at upper ends of the first and second shafts **102e**, **102f**, respectively.

The assembled control system **100** further includes an inner bearing element **120a** which in a further step after step 4) is moved in the assembly direction D to extend through the second shaft unit **102d** and through the connection hole **116b** of the housing frame **116a**. In the assembled control system **100**, the inner bearing element **120a** is rotationally fixed to the housing frame **116a**. The inner bearing element **120a** comprises an inner bearing surface **120c** (FIG. **3**) positioned farther from the common axis of rotation than the outer bearing surface **102i** of the first shaft **102e**, the inner bearing surface **120c** abutting the outer bearing surface **102i** so that rotation of the second control wheel unit **101b** is at least partly borne on the inner bearing element **120a**. The inner bearing element **120a** includes a cylindrical, tubular inner bearing sleeve having a wall. The inner bearing element **120a** separates rotation of the control wheel units **101a**, **101b** from each other.

The assembled control system further comprises an outer bearing element **120b** in one piece with the housing frame **116a**. The outer bearing element **120b** surrounds and defines the connection hole **116b** of the housing frame **116a**. The outer bearing element **120b** is tubular and cylindrical and comprises a wall which extends away from the housing frame **116a** in a direction towards the first control wheel unit

17

101a. The outer bearing element 120b comprises an inner bearing surface 120d (FIG. 3) positioned farther from the common axis of rotation than the outer bearing surface 102j of the second shaft 102f; the inner bearing surface 102j abutting the outer bearing surface 102j so that rotation of the second control wheel unit 101b is at least partly borne on the outer bearing element 120b. The outer bearing element 120b is an outer bearing sleeve that encompasses at least part of the second shaft 102e.

The control system 100 further includes a center shaft 103 which, in a further step after step 5), is moved in the assembly direction D to extend through the connection hole 116b of the housing frame 116a, the first and second shaft units 102c, 102d, and the center openings of the first and second control wheel units 101a, 101b. The center shaft 103 comprises a connector frame or center shaft frame 115, and the connector frame 115 is fixed by means of pins (not shown) to the housing frame 116a via the inner bearing element frame 121, see below, after insertion of the center shaft 103 so that the center shaft 103 is fixed to the housing frame 116a (not shown). The connector frame 115 extends radially from a shaft part 103a of the center shaft 103 and is positioned within the handle housing 116a in the assembled endoscope 1, see FIG. 3. The connector frame 115 includes a housing 115a including a flange 115b fixed to the housing frame. The housing 115a includes an indentation 115c that operates as stop or stop surface when it contacts a corresponding stop surface 102m on a longitudinally protruding portion of the first shaft unit 102c as it rotates. The protruding portion has an arcuate shape and its length determines the angle of rotation of the first shaft unit 102c.

Similarly, the inner bearing element 120a includes an inner bearing element frame 121 that extends radially from a sleeve part 120e thereof and is positioned within the handle housing 116 in the assembled endoscope 1. The inner bearing element frame 121 is directly fixed to the housing frame 116a, the center shaft frame 115 being directly fixed to the inner bearing element frame 121 by means of screws (not shown) so as to be indirectly fixed to the housing frame 116a. The screws are inserted into screw holes, one of these being designated 122 in FIG. 4.

When the center shaft 103 has been positioned, a cap 105a is moved opposite to the assembly direction D to be attached to a tip end 103b of the center shaft 103 by a snap connection 103c which is provided in a manner similar to the first and second snap connections 112, 113. Accordingly, the cap 105a includes two resilient and pushable connection parts 103d, whereas the tip end 103b includes associated two connection parts taking the form of recesses 103e. This snap engagement 103c is similarly be activated during or at the end of the insertion of the center shaft 103 into the control system 100. The cap 105a covers and attaches a first multi-disc brake 110a of the first control wheel unit 101a, see further below. The brake 110a is encased within a spacing defined by interior surfaces of the first wheel handle 101c. Before insertion of the brake 110a and subsequent positioning of the cap 105a, i.e. after steps 1) to 5), the assembled parts of the control system 100 are removed from the jig 123 and the assembled parts of the control system 100 are flipped around, i.e. turned 180 degrees, after which the first brake 110a is assembled and positioned to form part of the first control wheel unit 101a, after which, again, the cap 105a is positioned as described. Alternatively, the brake 110a and the cap 105a are mounted to form part of the first control wheel unit 101a before or during step 1). The cap 105a includes a brake knob 104a projecting in the assembly direction and upon rotation of which the brake 110a is

18

activated to brake rotation of the first control wheel unit 101a and, thus, first shaft unit 102d.

In the assembled control system 100, a brake handle 104b for activation of a similar, second multi-disc brake 110b, which brakes the second control wheel unit 101b in a similar manner, is attached to the housing frame 116a. During the method of assembly, when the second control wheel unit 101b has been positioned in step 2), before step 3), the brake handle 104b is moved in the assembly direction to be positioned on the second control wheel unit 101b.

In step 1), the first control wheel unit is positioned in and at the bottom of a correspondingly shaped part of a jig depression 124 of the jig 123 shown in FIG. 11, the jig 123 holding the first control wheel unit 101a in a fixed position. In step 2), the second control wheel unit is also positioned and held in a correspondingly shaped part of the jig depression 124. In step 3), the housing frame is also positioned and held in a correspondingly shaped part of the jig depression 124. The jig 123 includes a jig block 125 with in which the jig depression 123 is shaped to provide a positive engagement with the first control wheel unit 101a, especially the first wheel handle 101c. Similarly, the jig depression 124 is shaped to also provide a positive engagement with the second control wheel unit 101b, especially the second wheel handle 101d, when the second control wheel unit 101b is positioned on the first control wheel unit 101a in the jig 123. Similarly, the jig depression 124 is shaped to also provide a positive engagement with the housing frame 116a when the housing frame 116a is positioned on the second control wheel unit 101b in the jig 123. Hereby, all of these elements can be maintained in relative positions until the snap connections 112, 113 are engaged in steps 4) and 5).

After steps 1) to 5), the assembled control system 100 is removed from the jig 123, and further assembly occur as described. This involves assembly of the handle housing 116.

Referring to FIGS. 8 to 10, the second snap connection 113 axially and rotationally fixes the second shaft unit 102d to the second control wheel unit 101b. The second snap connection 103 includes two connection parts 113a in the form of resilient projections or pins of the second control wheel unit 101b which interlock with two associated connection parts 113b, embodied by recesses, of the second shaft unit 102d. These connection parts 113a, 113b are thus mutually engaging. During the movement of the second shaft unit 102d, the connection parts 113a are pushed in a radial direction (inwardly) and, then, when the second shaft unit 102d is further moved or inserted, resiliently snap back to engage with the associated connection parts 113b. The connection parts 113b includes a ramp or inclined surface 113c, which forces the pushable connection parts 113a inwardly in a radial direction during the movement of the second shaft unit 102d. Accordingly, the pushable connection parts 113a include a barb surface 113d which during the snap moves into engagement with an associated barb surface 113e of the other connection parts 113b to secure the position of the second shaft unit 102d to the second control wheel unit 101b. The pushable connection parts 113a are included in the second central part 102h, and the associated connection parts 113b are included in the second shaft unit 102d. The pushable connection parts 103a may each be provided by two axially extending slots defining a resilient, axially moveable pin between them, as in conventional snap fittings.

As described, the second snap connection 113 includes a barb surface 113d which during the snap moves into engagement with an associated barb surface 113e to secure the

19

position of the second shaft unit **102d** relative to the second control wheel unit **101b**. As shown, opposed walls, each comprising an inwardly projecting angled portion extending from a vertical portion below it, are provided on the second shaft unit **102d**. Each of the inwardly projecting angled portions includes an inclined surface **113c** and ends at the barb surface **113e**. In alternative embodiments, the inwardly projecting angled portion can be resilient and the connection parts **113a** can be of various rigidities, since the inwardly projecting angled portions can flex outwardly to allow the two connection parts **113a** to move past them into the cavities **113b**. Thus, the second snap connection **113** is made by opposing barb surfaces, at least one of which is provided by a radially resilient part.

Referring to FIGS. 5 to 7, like the second snap connection **113**, the first snap connection **112** axially and rotationally fixes the first shaft unit **102c** to the first control wheel unit **101a**. And the first snap connection **112** includes two connection parts **112a** of the first shaft unit **102c** interlocking with two associated connection parts **112b** (best seen in FIG. 7) of the first control wheel unit **101a**. These associated connection parts **112a**, **112b** mutually engage. During the movement of the first shaft unit **102c**, the connection parts **112a** are pushed inwardly in the radial direction by the wheel sleeve **102g** and, then, when the first shaft unit **102c** is further moved and inserted, resiliently snap back to engage the associated connection part **112b**. And again, this is achieved by the connection parts **112b** including a ramp or inclined surface **112c**, which forces the pushable connection parts **112a** in the radial direction during the movement of the first shaft unit **102c**. Accordingly, the pushable connection parts **112a** include a barb surface **112d** which during the snap moves into engagement with an associated, opposed barb surface **112e** of the connection parts **112b** to secure the position of the first shaft unit **102c** to the first control wheel unit **101a**. The pushable connection parts **112a** are included in the first shaft unit **102c**, in the first shaft **102e** thereof, and the associated connection parts **112b** are included in the first control wheel unit **101a**. The pushable connection parts **112a** may be provided by two axially extending slots defining a resilient, axially moveable pin between them.

As described, the first snap connection **112** includes a barb surface **112d** which during the snap moves into engagement with an associated barb surface **112e** to secure the position of the first unit **102c** relative to the second control wheel unit **101a**. In alternative embodiments, the connection parts **112b** do not need to include a ramp or inclined surface **112c**. The pushable connection parts **112a** each includes a ramp or inclined surface at the distal end thereof, which can force the pushable connection parts **112a** inwardly in the radial direction during the movement of the first shaft unit **102c** when the ramps contact the inner surface of the wheel sleeve **102g**. Thus, the first snap connection **112** is made by opposing barb surfaces, at least one of which is provided by a radially resilient part.

Hereby, the first and second snap connections **112**, **113**, when engaged, prevent movement of the first and second control wheels **101a**, **101b**, the housing frame **116a**, and the first and second shaft units **102c**, **102d** relative to each other in the assembly direction D. Accordingly, the barb surfaces **112d**, **112e**; **113d**, **113e** face in the axial direction and are mutually opposed to each other during and after assembly. Barbs of the connection parts **112a**, **113a** are provided as projections extending in the radial direction.

The assembly direction D is parallel with the axes of rotation of the first and second control wheel units **101a**, **101b**.

20

As shown in FIG. 3, in the assembled control system **100**, the first and second control wheel units **101a**, **101b** house the associated brakes **110a**, **110b**. Activation of each of the brakes **110a**, **110b** moves the brake from a released position to a braking position. A brake force of the brake **110a**, **110b** in the braking position brakes rotation of the associated control wheel unit **101a**, **101b**, respectively. The brake force is released in the released position. The brake **110b** is assembled in the second wheel handle before step 1). The brakes **110a**, **110b** each includes a stack of brake discs and a helical spring **117a**, **117b**, respectively.

After assembly of the control system **100** and then the handle **2**, the control system **100** forms part of the handle **2**.

Rotation of the control wheel units **101a**, **101b** occurs relative to the housing frame **116a** during the bending operation.

The handle housing **116** takes the form of a housing shell.

The first and/or second wire drums **102a**, **102b** are pulleys. In the assembled endoscope **1**, the first and second steering wires are attached to the wire drums **102a**, **102b** to be woundable on these, respectively.

Any one or more of or all steps 1) to 5) can be carried out manually and/or automatically. Any one or more or all of the further steps of assembly may also be carried out manually and/or automatically.

In each of steps 2) to 5), all already positioned parts of the control system **100** remain in the held position.

The snap connection parts **112a**, **112b**; **113b**, **113a** are provided in one piece with the first shaft unit **102c** and the first control wheel unit **101a**; the second shaft unit **102d** and the second control wheel unit **101b**, respectively.

The foregoing aspects are further embodied in the following exemplary items:

Item 1. A method of assembly of an endoscope control system, the endoscope control system being for performing a bending operation in a disposable insertion endoscope, wherein the endoscope control system comprises: a housing frame for forming or for forming part of an endoscope handle housing, the housing frame comprising a connection hole; a first control wheel unit comprising a first wheel handle; a second control wheel unit comprising a second wheel handle; a first shaft unit, the first shaft unit comprising a first wire drum and a first shaft, the first shaft connecting the first control wheel unit to the first wire drum, the first wire drum being for connection to a first steering wire of the endoscope, whereby rotation of the first wheel handle relative to the housing frame about an axis of rotation controls the bending operation in a first dimension; a second shaft unit, the second shaft unit comprising a second wire drum and a second shaft, the second shaft connecting the second control wheel unit to the second wire drum, the second wire drum being for connection to a second steering wire of the endoscope, whereby rotation of the second wheel handle relative to the housing frame about an axis of rotation controls the bending operation in a second dimension; and wherein the method of assembly comprises, in sequence, the steps of: 1) holding the first control wheel unit in a position; 2) moving the second control wheel unit in an assembly direction to position the second control wheel unit on the first control wheel unit; 3) moving the housing frame in the assembly direction to position the housing frame on the second control wheel unit; 4) moving the second shaft unit in the assembly direction so that the second shaft is positioned to extend through the connection hole of the housing frame and snaps into engagement with the second control wheel unit by means of a second snap connection between the second shaft and the second control wheel unit; and 5)

21

moving the first shaft unit in the assembly direction so that the first shaft is positioned to extend through the connection hole of the housing frame and through the second shaft and snaps into engagement with the first control wheel unit by means of a first snap connection between the first shaft and the first control wheel unit; whereby the first and second control wheel units, the housing frame, and the first and second shaft units are maintained in position relative to each other in the assembly direction by means of the first and second snap connections.

Item 2. The method of item 1, wherein all the movements of steps 2) to 5) are carried out from one side only.

Item 3. The method of item 1 or 2, wherein, in each of steps 2) to 5), all already positioned parts of the control system to be assembled remain in a held position.

Item 4. The method of any one of the previous items, wherein only the movement carried out in steps 4) and 5) affect the attachment of the parts assembled in steps 1) to 5) to each other.

Item 5. The method of any one of the previous items, wherein, during the sequence of steps 1) to 5), no separate locking device is applied to attach the parts to each other.

Item 6. The method of any one of the previous items, wherein the first and second snap connections are provided only by snap connection parts provided in one piece with the first shaft unit, the first control wheel unit, the second shaft unit, and the second control wheel unit, respectively.

Item 7. The method of any one of the previous items, wherein the first snap connection comprises at least one primary connector part forming part of the first shaft unit and at least one secondary connector part forming part of the first control wheel unit, the primary and secondary connector parts in step 5) snapping directly onto each other to form the first snap connection; and wherein the second snap connection comprises at least one tertiary connector part forming part of the second shaft unit and at least one quaternary connector part forming part of the second control wheel unit, the tertiary and quaternary connector parts in step 4) snapping directly onto each other to form the second snap connection.

Item 8. The method of any one of the previous items, wherein no further steps are involved in the method during the sequence of steps 1) to 5).

Item 9. The method of any one of the previous items wherein, before step 1), the first wheel is positioned in a jig, the first wheel being held in the jig in step 1) and during the sequence of steps 2) to 5).

Item 10. An endoscope control system for performing a bending operation in a disposable insertion endoscope, wherein the endoscope control system comprises: a housing frame for forming or for forming part of an endoscope handle housing, the housing frame comprising a connection hole; a first control wheel unit comprising a first wheel handle; a second control wheel unit comprising a second wheel handle; a second shaft unit, the second shaft unit comprising a second wire drum and a second shaft, the second shaft connecting the second control wheel unit to the second wire drum, the second wire drum being for connection to a second steering wire of the endoscope, whereby rotation of the second wheel handle relative to the housing frame about an axis of rotation controls the bending operation in a second dimension; wherein the second control wheel is positioned between the first control wheel and the housing frame, the second shaft extends through the connection hole of the housing frame, and the first shaft extends through the connection hole of the housing frame and through the second shaft; and wherein the first shaft is

22

connected to the first control wheel unit by means of a first snap connection between the first shaft and the first control wheel unit, and the second shaft is connected to the second control wheel unit by a second snap connection between the second shaft and the second control wheel unit whereby the first and second snap connections attach the first and second control wheels, the housing frame, and the first and second shaft units to each other.

Item 11. The control system according to item 10, wherein the first and second snap connections maintain attachment between the first and second control wheels, the housing frame, and the first and second shaft units if all other parts are removed.

Item 12. The control system according to item 10 or 11, wherein the first and second snap connections are provided by snap connection parts provided in one piece with the first shaft unit, the first control wheel unit, the second shaft unit, and the second control wheel unit, respectively.

Item 13. The endoscope control system according to any one of items 10 to 12, wherein the endoscope control system has been assembled according to the method of any one of items 1 to 9.

Item 14. An endoscope comprising the control system assembled according to any one of items 1 to 9 and/or comprising the control system according to any one of items 10 to 13.

Item 15. The endoscope according to item 14, further comprising the first and second steering wires and a distal tip or tip part that comprises a bending section connected to the first and second steering wires so that the control system can activate the bending operation of the bending section via the steering wires.

Item 16. The endoscope according to item 14, wherein the first shaft includes a bearing surface, the second shaft includes an outer bearing surface positioned farther from the axis of rotation than the bearing surface of the first shaft, and the handle housing includes an outer bearing element comprising an inner bearing surface positioned farther from the axis of rotation than the outer bearing surface of the second shaft, the inner bearing surface of the outer bearing element abutting the outer bearing surface of the second shaft so that rotation of the second control wheel is at least partly borne on the outer bearing element.

Item 17. The endoscope of item 16, further comprising an inner bearing element framed disposed at least in part between the first shaft and the second shaft, wherein the inner bearing element frame axially separates the first and second control wheel units from each other.

LIST OF REFERENCE SIGNS

- 1 Endoscope
- 2 Endoscope handle
- 3 Elongated insertion tube
- 3a Proximal end of insertion tube
- 3b Distal end of insertion tube
- 4 Tip
- 5 Bending section
- 6 Suction connector
- 7 Working channel port
- 100 Endoscope control system
- 101 First control wheel
- 101a First control wheel unit
- 101b Second control wheel unit
- 101c First wheel handle
- 101d Second wheel handle
- 102 Second control wheel

23

102a First wire drum
 102b Second wire drum
 102c First shaft unit
 102d Second shaft unit
 102e First shaft
 102f Second shaft
 102g First central part
 102h Second central part
 102i Bearing surface of the first shaft
 102j Bearing surface of the second shaft
 102k First steering wire
 102l Second steering wire
 103 Center shaft
 103a Shaft part
 103b Tip end
 103c Snap connection
 103d Connection parts
 103e Recesses
 104a Brake knob
 104b Brake handle
 105a Cap
 110a First multi-disc brake
 110b Second multi-disc brake
 111 Stack of brake discs
 112 First snap connection
 112a Snap connection parts
 112b Snap connection parts
 112c Ramps
 112d Barb surface
 112e Barb surface
 113 Second snap connection
 113a Snap connection parts
 113b Snap connection parts
 113c Ramps
 113d Barb surface
 113e Barb surface
 115 Connector/center shaft frame
 116 Handle housing
 116a Housing frame
 116b Connection hole
 117a Spring
 117b Spring
 120a Inner bearing element
 120b Outer bearing element
 120c Inner bearing surface
 120d Inner bearing surface
 120e Sleeve part
 121 Inner bearing element frame
 122 Screw hole
 123 Jig
 124 Jig depression
 125 Jig block
 D Assembly direction

I claim:

1. A method of assembly of an endoscope, the method comprising:

providing a housing frame including a first shell part and a second shell part, the first shell part comprising an outer bearing sleeve, the outer bearing sleeve including an inner bearing surface and a connection hole;
 providing a first shaft unit including a first shaft;
 providing a second shaft unit including a second shaft having a central hole sized and shaped to receive therethrough the first shaft and defining an axial direction;

24

moving a second control wheel unit in the axial direction to position the second control wheel unit on a first control wheel unit;
 moving the first shell part in the axial direction to position the first shell part on the second control wheel unit;
 moving the second shaft unit in the axial direction to position the second shaft through the connection hole of the first shell part until the second shaft snaps into engagement with the second control wheel unit forming a second snap connection; and
 moving the first shaft unit in the axial direction to position the first shaft through the central hole of the second shaft and through the connection hole until the first shaft snaps into engagement with the first control wheel unit forming a first snap connection;
 wherein the first snap connection and the second snap connection maintain the first and second control wheel units, the housing frame, and the first and second shaft units assembled relative to each other,
 wherein the first snap connection and the second snap connection are provided only by snap connection parts provided in one piece with each of the first shaft unit, the first control wheel unit, the second shaft unit, and the second control wheel unit, respectively,
 wherein the first snap connection comprises at least one primary connector part forming part of the first shaft unit and at least one secondary connector part forming part of the first control wheel unit, the at least one primary connector part and at least one secondary connector part snapping directly onto each other to form the first snap connection, and
 wherein the second snap connection comprises at least one tertiary connector part forming part of the second shaft unit and at least one quaternary connector part forming part of the second control wheel unit, the at least one tertiary connector part and at least one quaternary connector part snapping directly onto each other to form the second snap connection.
 2. The method of claim 1, the method further comprising: providing an inner bearing element sleeve, wherein the second snap connection is positioned intermediate the inner bearing element sleeve and the outer bearing sleeve.
 3. The method of claim 1, the method further comprising: moving an inner bearing element sleeve in the axial direction to position the inner bearing element sleeve through the connection hole and the second shaft, wherein the second snap connection is positioned intermediate the inner bearing element sleeve and the outer bearing sleeve.
 4. The method of claim 1, wherein moving the second control wheel unit, the housing frame, the second shaft unit and the first shaft unit comprise moving the second control wheel unit, the housing frame, the second shaft unit and the first shaft unit toward the first control wheel unit.
 5. The method of claim 4, wherein the first control wheel unit, the second control wheel unit, the housing frame, the second shaft unit and the first shaft unit are components of a stack, and wherein the method comprises stacking the components in order to form the stack.
 6. The method of claim 5, wherein stacking the components in order further comprises holding a portion of the stack while moving the components onto the portion of the stack being held.
 7. The method of claim 6, wherein the first control wheel unit comprises a first wheel handle, and wherein the method further comprises holding the first wheel handle in a jig, and then stacking the components onto the first control wheel unit.

8. The method of claim 5, wherein the method further comprises, after moving the second shaft unit, moving an inner bearing element in the axial direction to position the inner bearing element through the connection hole of the housing frame.

5

9. The method of claim 8, wherein the method further comprises, after moving the first shaft unit, moving a center shaft in the axial direction to position the center shaft through the first shaft and moving a cap in the axial direction toward the first control wheel unit to form a snap connection between a tip end of the center shaft and the cap to secure the center shaft in place.

10

10. The method of claim 1, wherein moving the second control wheel unit, the housing frame, the second shaft unit and the first shaft unit are carried out from one side only.

15

11. The method of claim 10, wherein the second control wheel unit, the housing frame, the second shaft unit and the first shaft unit remain in a held position after each has been positioned.

12. The method of claim 1, wherein forming the first snap connection and the second snap connection are sufficient to attach the first control wheel unit, the second control wheel unit, the housing frame, the second shaft unit and the first shaft unit to each other.

20

13. The method of claim 12, wherein, during the moving of the second control wheel unit, the housing frame, the second shaft unit and the first shaft unit, no separate locking device is applied to attach the first control wheel unit, the second control wheel unit, the housing frame, the second shaft unit and the first shaft unit to each other.

25

30

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